Problem Definition Add dynamic UI components such as collapsible FAQ, modal popup, image/content slider, notification banner, hamburger menu, and light/dark theme switcher using JavaScript.:-

CODE:-

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Dynamic UI Practical</title>

    <style>
        * {
            box-sizing: border-box;
        }

        body {
            font-family: Arial, sans-serif;
            margin: 0;
            padding: 0;
            background-color: #f4f6f8;
```

```
    color: #222;
    transition: 0.3s;
}

header {
    background-color: #222;
    color: white;
    padding: 15px 30px;
    display: flex;
    align-items: center;
    gap: 15px;
}

header h2 {
    margin-right: auto;
}

button {
    padding: 9px 15px;
    border: none;
    border-radius: 5px;
    cursor: pointer;
    background-color: #007bff;
```

```
    color: white;
    font-size: 14px;
}

button:hover {
    background-color: #0056b3;
}

nav {
    display: none;
}

nav a {
    color: white;
    text-decoration: none;
    margin: 0 10px;
}

nav a:hover {
    text-decoration: underline;
}

#notification {
```

```
    background-color: #dff0d8;
    color: #2e6b2e;
    padding: 15px 25px;
    margin: 20px auto;
    width: 90%;
    border-radius: 8px;
}

#notification button {
    float: right;
    background-color: transparent;
    color: #333;
    font-size: 20px;
    padding: 0 5px;
}

main {
    width: 90%;
    max-width: 900px;
    margin: auto;
}

h1 {
```

```
    text-align: center;
    margin: 30px 0;
}

h2 {
    margin-top: 30px;
}

.question {
    background-color: white;
    padding: 15px;
    margin-top: 10px;
    border-radius: 6px;
    cursor: pointer;
    border: 1px solid #ddd;
}

.question:hover {
    background-color: #e9f2ff;
}

.answer {
    display: none;
```

```
    background-color: #f8f9fa;
    padding: 15px;
    border-left: 4px solid #007bff;
    margin-bottom: 10px;
}

#popup {
    display: none;
    position: fixed;
    top: 50%;
    left: 50%;
    transform: translate(-50%, -50%);
    background-color: white;
    padding: 25px;
    width: 350px;
    border-radius: 10px;
    box-shadow: 0 5px 20px rgba(0, 0, 0, 0.3);
    z-index: 2;
    text-align: center;
}

#overlay {
    display: none;
```

```
    position: fixed;
    top: 0;
    left: 0;
    width: 100%;
    height: 100%;
    background-color: rgba(0, 0, 0, 0.6);
    z-index: 1;
}

.slider {
    text-align: center;
    margin-top: 20px;
}

#slideImage {
    width: 600px;
    max-width: 100%;
    height: 300px;
    object-fit: cover;
    border-radius: 10px;
    box-shadow: 0 3px 10px rgba(0, 0, 0, 0.2);
}
```

```
#slideText {
    margin: 15px;
}

.slider button {
    margin: 5px;
}

.dark {
    background-color: #111;
    color: white;
}

.dark .question {
    background-color: #222;
    color: white;
    border-color: #444;
}

.dark .answer {
    background-color: #222;
    color: white;
}
```

```
.dark #popup {
    background-color: #222;
    color: white;
}

@media (max-width: 600px) {
    header {
        flex-wrap: wrap;
    }

    header h2 {
        width: 100%;
    }

    nav {
        width: 100%;
    }

    nav a {
        display: block;
        margin: 8px 0;
    }
```

```
        }
    </style>
</head>

<body>

    <header>

        <h2>My Portal</h2>

        <button onclick="toggleMenu()">☰ Menu</button>

        <nav id="menu">
            <a href="#">Home</a>
            <a href="#">About</a>
            <a href="#">Contact</a>
        </nav>

        <button onclick="toggleTheme()">🌙 Theme</button>

    </header>
```

```
<div id="notification">
    Welcome to our Dynamic UI Portal!

    <button onclick="closeNotification()">×</button>
</div>


<main>

    <h1>Dynamic UI Practical</h1>


    <h2>1. Collapsible FAQ</h2>

    <div class="question" onclick="toggleAnswer(0)">
        <b>What is this portal?</b>
    </div>

    <div class="answer">
        This portal helps users interact with various UI elements dynamically.
    </div>
```

```
<div class="question" onclick="toggleAnswer(1)">
  <b>How to use the slider?</b>
</div>

<div class="answer">
  Use the buttons below the slider to view different slides.
</div>

<div class="question" onclick="toggleAnswer(2)">
  <b>What is JavaScript?</b>
</div>

<div class="answer">
  JavaScript is used to make web pages interactive and dynamic.
</div>

<h2>2. Modal Popup</h2>

<button onclick="showPopup()">
```

```
    Show Popup
</button>


<div id="overlay"></div>


<div id="popup">

    <h3>Popup Message</h3>

    <p>This is a dynamic popup message!</p>

    <button onclick="closePopup()">Close</button>

</div>


<h2>3. Image / Content Slider</h2>

<div class="slider">

    <img id="slideImage"
```

```
            src="https://picsum.photos/id/1015/600/300"
            alt="Slider Image">

        <h3 id="slideText">Beautiful Mountain</h3>

        <button onclick="previousSlide()">Previous</button>

        <button onclick="nextSlide()">Next</button>

    </div>

</main>
<script>

    function toggleAnswer(index) {

        let answers = document.querySelectorAll(".answer");

        if (answers[index].style.display == "block") {

            answers[index].style.display = "none";

        }
```

```
        else {

            answers[index].style.display = "block";

        }

    }


    function showPopup() {

        document.getElementById("popup").style.display =
"block";

        document.getElementById("overlay").style.display =
"block";

    }


    function closePopup() {

        document.getElementById("popup").style.display =
"none";
```

```
        document.getElementById("overlay").style.display =
"none";

    }


    let images = [

        "https://picsum.photos/id/1015/600/300",

        "https://picsum.photos/id/1025/600/300",

        "https://picsum.photos/id/1043/600/300"

    ];


    let texts = [

        "Beautiful Mountain",

        "Cute Dog",
```

```
    "Beautiful Nature"

];


let currentSlide = 0;


function showSlide() {

    document.getElementById("slideImage").src =
        images[currentSlide];

    document.getElementById("slideText").innerHTML =
        texts[currentSlide];

}


function nextSlide() {

    currentSlide++;
```

```
    if (currentSlide >= images.length) {
        currentSlide = 0;
    }
    showSlide();
}

function previousSlide() {
    currentSlide--;
    if (currentSlide < 0) {
        currentSlide = images.length - 1;
    }
    showSlide();
```

```
    }


    function closeNotification() {

        document.getElementById("notification").style.display
= "none";

    }


    function toggleMenu() {

        let menu = document.getElementById("menu");

        if (menu.style.display == "block") {

            menu.style.display = "none";

        }
        else {
```

```
            menu.style.display = "block";

        }

    }

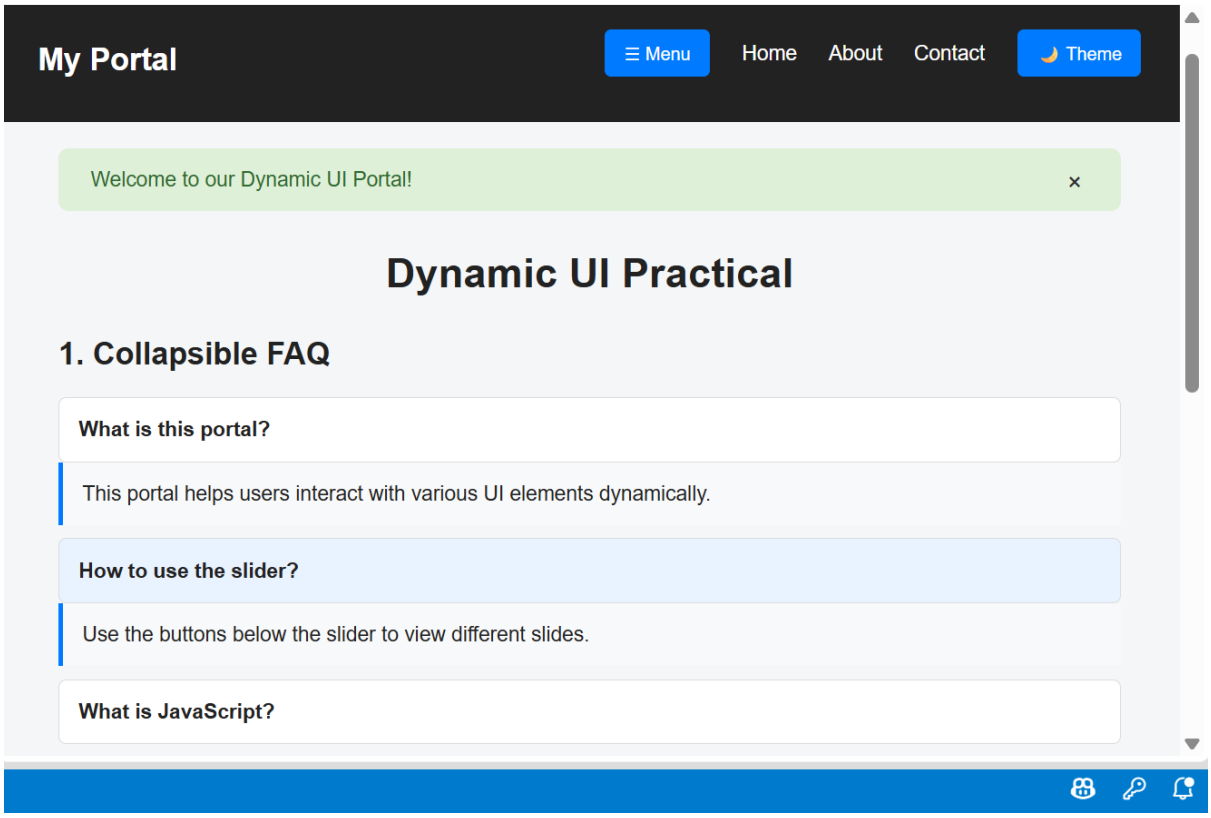
    function toggleTheme() {

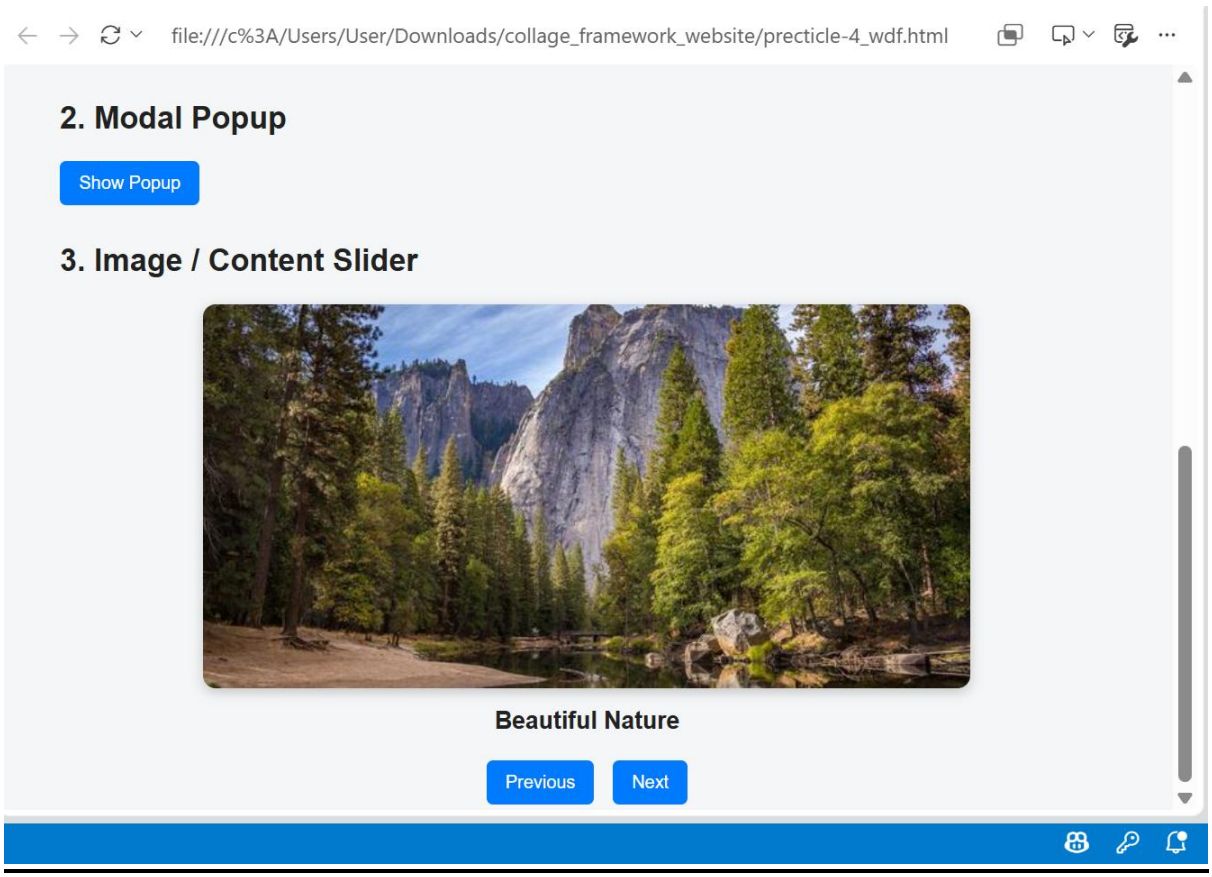
        document.body.classList.toggle("dark");

    }

  </script>
</body>
</html>
```

Output:-


My Portal
☰ Menu
Home
About
Contact
🌙 Theme
Welcome to our Dynamic UI Portal!
×
Dynamic UI Practical
1. Collapsible FAQ
What is this portal?
This portal helps users interact with various UI elements dynamically.
How to use the slider?
Use the buttons below the slider to view different slides.
What is JavaScript?


file:///c%3A/Users/User/Downloads/collage_framework_website/precticle-4_wdf.html
2. Modal Popup
Show Popup
3. Image / Content Slider
Beautiful Nature
Previous
Next

Questions:-

1. How are DOM elements selected and modified?
DOM elements are selected using methods like document.getElementById() and document.querySelectorAll(). Their properties such as style.display, src, innerHTML, and classList are then modified to create dynamic behavior.

2. Are event listeners attached correctly?
Yes. Event handlers are attached using onclick attributes, such as onclick="toggleMenu()", onclick="showPopup()", and onclick="nextSlide()". They correctly trigger the required JavaScript functions.

3. Is localStorage used for remembering theme choice?
No. In the current code, localStorage is not used. The theme changes only while the current page is open. After refreshing the page, the theme returns to the default.

4. Does interactivity improve usability without breaking accessibility?
Yes, generally. The FAQ, popup, menu, notification, and slider make the page more interactive and user-friendly. However, accessibility can be improved by using proper semantic elements, keyboard-accessible controls, ARIA attributes where needed, and avoiding reliance only on mouse clicks.

Tools/Technology to be Used :-

JavaScript ES6+, HTML5, CSS3, browser dev tools

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